Chem 0320 – Organic Chemistry 2
EXAM 1 (Wednesday)
Spring 2024 (2244)
Dr. Peter Bell

First two (**2**) letters of *last* name:

Name (printed legibly): ____

Signature: ____

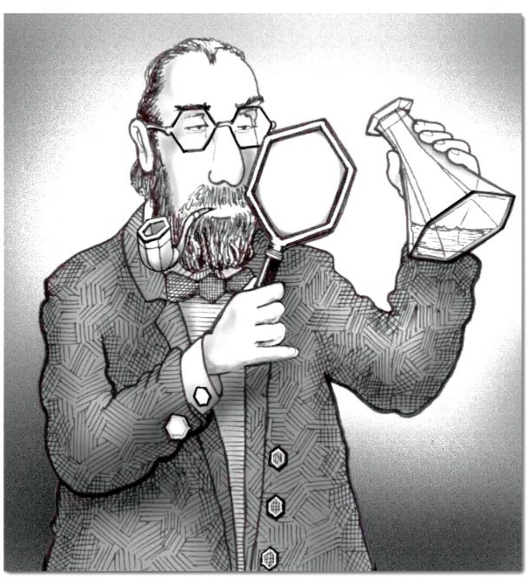
Student ID #: ___ ___ ___ ___ ___ ___ ___

DO NOT OPEN this exam until instructed and the entire class begins.
Opening this prior is a violation of the Academic Integrity Code
as is continuing to write after time is called.
You may tear the periodic table/scratch paper off the back while waiting to begin.

We've covered a variety of chemistry where electrons work together beyond a single bonding interaction between atoms. Likewise I hope you have taken time to work together with classmates, peer-leaders, and me. For now, working together has come to a pause for this exam. Now as individuals – work carefully. Work completely. And work concisely using drawings!

Pete

GREAT EVENTS IN CHEMISTRY

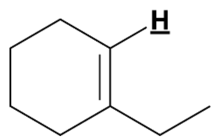


1865: Kekulé, moments before his brilliant insight into the structure of benzene.

page 2	______________	(18)
page 3	______________	(10)
page 4	______________	(16)
page 5	______________	(16)
page 6	______________	(16)
page 7	______________	(16)
page 8	______________	(8)
TOTAL	______________	(100)

1. Draw a compound containing an aromatic functional group and no others. (4)

2. Give the pKa of the bold, underlined hydrogen in the following compound. (4)

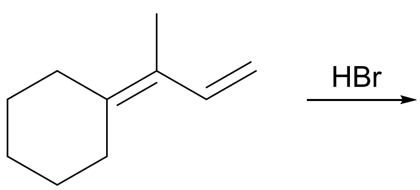


pka

3. Complete the following for the addition of HBr to the compound below. (10)

   a) (3 pts each) Draw the products of 1,2- and 1,4-addition.

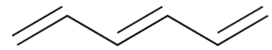
   b) (2 pts each) Fill in the bubbles indicating which is the kinetic product or the thermodynamic product. If both are kinetic or both are thermodynamic, fill in the bubble below each for that label.



1,2 product	**1,4 product**
⬭ kinetic product ⬭ thermodynamic product	⬭ kinetic product ⬭ thermodynamic product

Scratch Space (will not be graded)

4. Consider the molecule 1,3,5-hexatriene. Answer the following. (10)



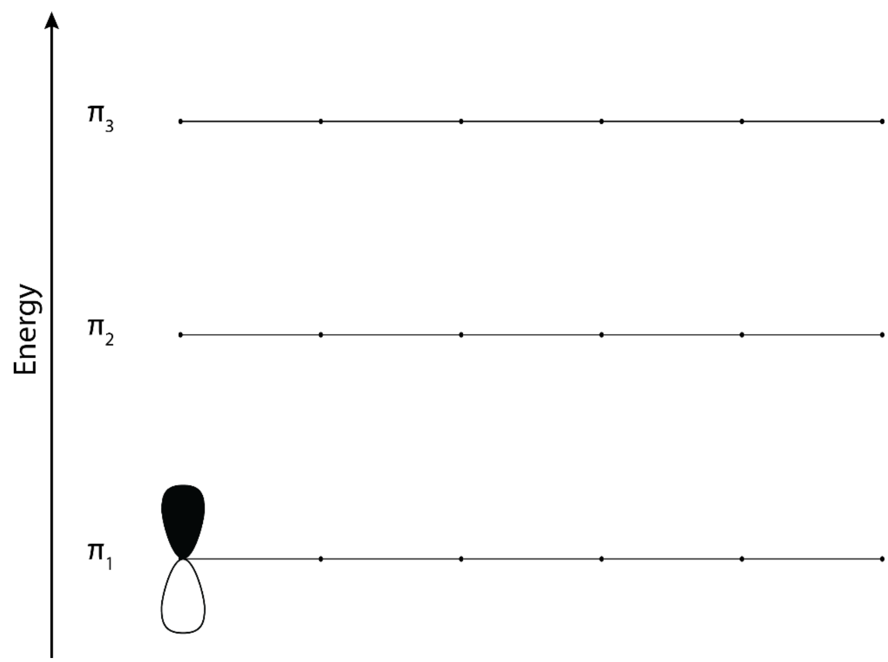
a) (3) How many π molecular orbitals are there? (How many can be described or drawn in an energy diagram?)

__________ total number of π molecular orbitals

b) (3) How does the presence of nodes change the energy of orbitals?

- ⬭ no change, just a product of combining orbitals
- ⬭ decrease
- ⬭ increase

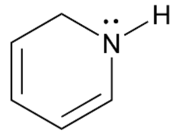
c) (4) In the space below, draw the three (3) ***lowest*** energy π molecular orbitals of 1,3,5-hexatriene. Shade the orbitals to indicate their phases. Indicate any nodes in the MOs with dashed lines.



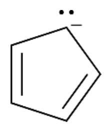
your initials
or signature

5. Identify the following as aromatic, non-aromatic, or anti-aromatic. (8)

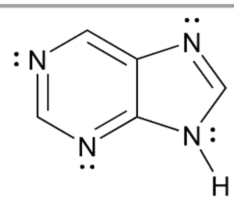
- aromatic
- anti-aromatic
- non-aromatic



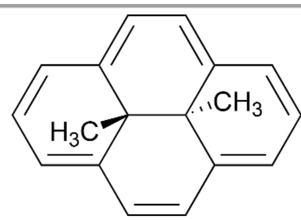
- aromatic
- anti-aromatic
- non-aromatic



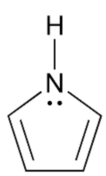
- aromatic
- anti-aromatic
- non-aromatic



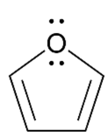
- aromatic
- anti-aromatic
- non-aromatic



6. Shown below are the structures of two aromatic compounds – pyrrole and furan. Using mostly drawings and some words, explain why pyrrole has a greater resonance stabilization energy (21 kcal/mol) than furan (16 kcal/mol). (8)



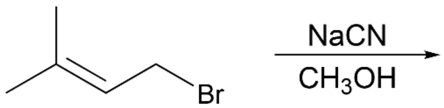
pyrrole

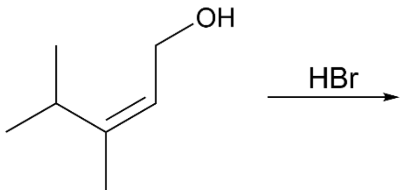


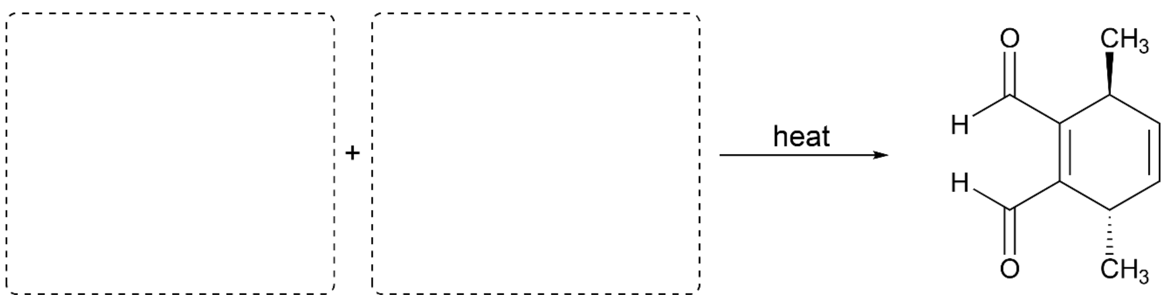
furan

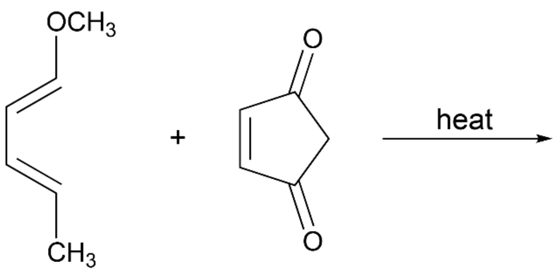
7. Complete the following reactions. In dashed boxes over arrows of reactions where products are shown, fill in the necessary conditions. Give the major product(s) of the other reactions. Be sure to show or otherwise indicate relevant stereochemical outcomes where appropriate. If no reaction takes place, write "no reaction" or "NR".

   Write out all reagents. For example, if the Bell Reagent was NaOEt, please write NaOEt or $NaOCH_2CH_3$ – not "Bell Reagent." (32)









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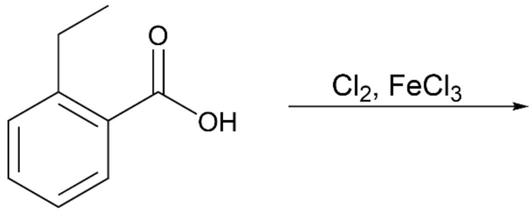
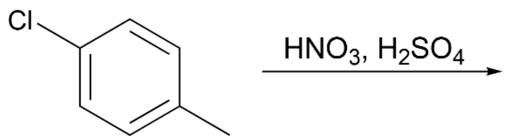
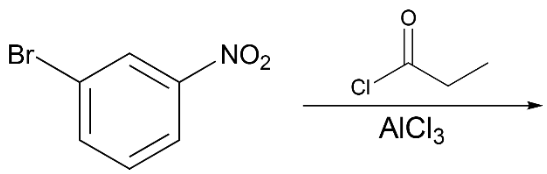
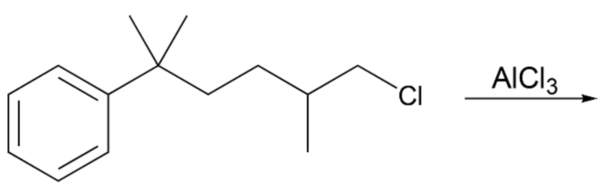




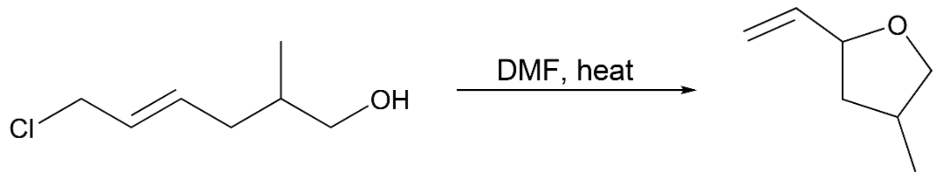




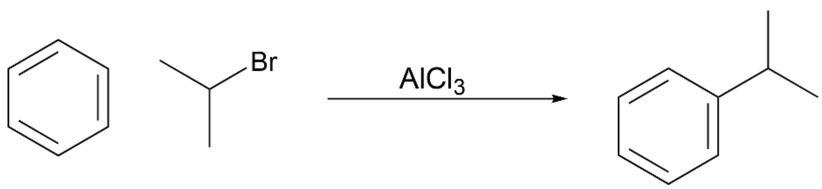
Hint: Watch for rearrangements!

8. Write out the complete, step-by-step mechanism using curved arrows for the following reactions. Be certain that there is no ambiguity where arrows begin or end and that all bonds formed or broken are explicitly shown.

a) (8)



b) (8)



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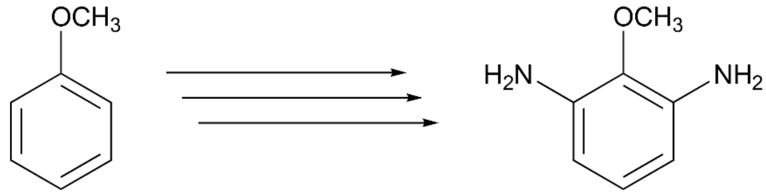
9. Propose a reasonable synthesis of the following tri-substituted benzene. (8)



Periodic Table of the Elements

1 IA 1A	2 IIA 2A	3 IIIB 3B	4 IVB 4B	5 VB 5B	6 VIB 6B	7 VIIB 7B	8	9 VIII 8	10	11 IB 1B	12 IIB 2B	13 IIIA 3A	14 IVA 4A	15 VA 5A	16 VIA 6A	17 VIIA 7A	18 VIIIA 8A
1 **H** 1.008																	2 **He** 4.003
3 **Li** 6.941	4 **Be** 9.012											5 **B** 10.811	6 **C** 12.011	7 **N** 14.007	8 **O** 15.999	9 **F** 18.998	10 **Ne** 20.180
11 **Na** 22.990	12 **Mg** 24.305											13 **Al** 26.982	14 **Si** 28.086	15 **P** 30.974	16 **S** 32.066	17 **Cl** 35.453	18 **Ar** 39.948
19 **K** 39.098	20 **Ca** 40.078	21 **Sc** 44.956	22 **Ti** 47.867	23 **V** 50.942	24 **Cr** 51.996	25 **Mn** 54.938	26 **Fe** 55.845	27 **Co** 58.933	28 **Ni** 58.693	29 **Cu** 63.546	30 **Zn** 65.38	31 **Ga** 69.723	32 **Ge** 72.631	33 **As** 74.922	34 **Se** 78.971	35 **Br** 79.904	36 **Kr** 83.798
37 **Rb** 85.468	38 **Sr** 87.62	39 **Y** 88.906	40 **Zr** 91.224	41 **Nb** 92.906	42 **Mo** 95.95	43 **Tc** 98.907	44 **Ru** 101.07	45 **Rh** 102.906	46 **Pd** 106.42	47 **Ag** 107.868	48 **Cd** 112.414	49 **In** 114.818	50 **Sn** 118.711	51 **Sb** 121.760	52 **Te** 127.6	53 **I** 126.904	54 **Xe** 131.294
55 **Cs** 132.905	56 **Ba** 137.328	57-71	72 **Hf** 178.49	73 **Ta** 180.948	74 **W** 183.84	75 **Re** 186.207	76 **Os** 190.23	77 **Ir** 192.217	78 **Pt** 195.085	79 **Au** 196.967	80 **Hg** 200.592	81 **Tl** 204.383	82 **Pb** 207.2	83 **Bi** 208.980	84 **Po** [208.982]	85 **At** 209.987	86 **Rn** 222.018
87 **Fr** 223.020	88 **Ra** 226.025	89-103	104 **Rf** [261]	105 **Db** [262]	106 **Sg** [266]	107 **Bh** [264]	108 **Hs** [269]	109 **Mt** [278]	110 **Ds** [281]	111 **Rg** [280]	112 **Cn** [285]	113 **Nh** [286]	114 **Fl** [289]	115 **Mc** [289]	116 **Lv** [293]	117 **Ts** [294]	118 **Og** [294]

57 **La** 138.905	58 **Ce** 140.116	59 **Pr** 140.908	60 **Nd** 144.243	61 **Pm** 144.913	62 **Sm** 150.36	63 **Eu** 151.964	64 **Gd** 157.25	65 **Tb** 158.925	66 **Dy** 162.500	67 **Ho** 164.930	68 **Er** 167.259	69 **Tm** 168.934	70 **Yb** 173.055	71 **Lu** 174.967
89 **Ac** 227.028	90 **Th** 232.038	91 **Pa** 231.036	92 **U** 238.029	93 **Np** 237.048	94 **Pu** 244.064	95 **Am** 243.061	96 **Cm** 247.070	97 **Bk** 247.070	98 **Cf** 251.080	99 **Es** [254]	100 **Fm** 257.095	101 **Md** 258.1	102 **No** 259.101	103 **Lr** [262]